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Artificial Intelligence

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INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Introduction To Artificial Intelligence

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INTRODUCTION

Briefly History, of Artificial Neural Networks (ANNs).

The history of Artificial Neural Networks (ANNs) is a fascinating journey that spans several decades, marked by periods of significant development and periods of relative stagnation the overview of the key

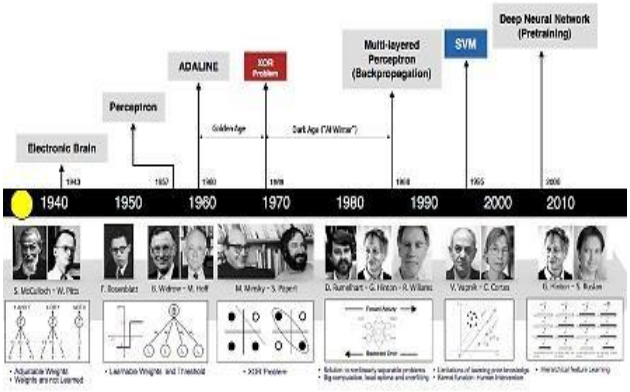

Figure 1 history of ANN

milestones in the history of ANNs are:

1. **1940s - 1950s: Early Beginnings**
   - The concept of artificial neurons and neural networks was inspired by the study of the human brain.
   - Warren McCulloch and Walter Pitts published a paper in 1943 introducing the first mathematical model of a neuron.
2. **1960s - 1970s: The First AI Winter**
   - Early enthusiasm for ANNs waned due to limitations in Perceptrons and a lack of computing power.

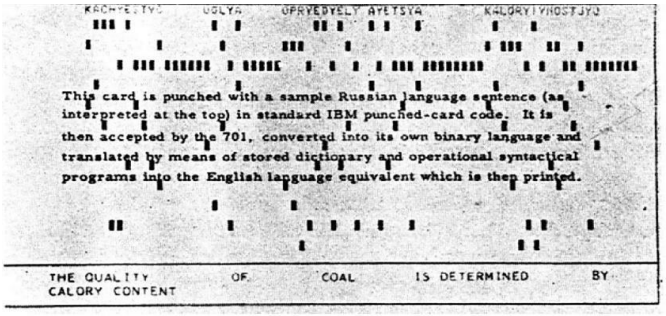

Figure 2 how ANN compiles information.

As shown in figure 2 above in 1955, Arthur Samuel wrote a program that could play checkers very well. A year later, it even appeared on television. It used a combination of a tree search with heuristics and learned weights. Samuel handcrafted the heuristics inspired by a book from checkers experts. He used a learning algorithm he called "temporal-difference learning" where the weights are adjusted using the "error" between the score initially calculated and the score after the search was completed.

- Marvin Minsky and Seymour Papert published a book in 1969 called

"Perceptrons," which highlighted the limitations of single-layer neural networks.

3. **1980s: Back propagation and Resurgence**
   - The development of the backpropagation algorithm by David Rumelhart, Geoffrey Hinton, and Ronald Williams in 1986 was a major breakthrough. It allowed for the training of multi-layer neural networks.
   - This period saw renewed interest in ANNs, leading to the first

successful applications in fields like speech recognition and image processing.

4. **1990s: Practical Applications**
    - ANNs began to be used in various applications, including financial modeling, hand-written character recognition, and speech recognition.
    - Support vector machines (SVMs) and other machine learning algorithms gained popularity, temporarily overshadowing ANNs.
5. **2000s: Deep Learning Emerges**
    - Deep Learning, which involves neural networks with multiple hidden layers (Deep Neural Networks, or DNNs), gained traction.
    - GPUs (Graphics Processing Units) became available for general-purpose computing, greatly accelerating the training of deep networks.
    - Geoff Hinton's work on Restricted Boltzmann Machines (RBMs) and deep belief networks contributed to the resurgence of deep learning.
6. **2010s: Deep Learning Dominance**
    - Deep learning achieved breakthroughs in various fields, including image recognition, natural language processing, and autonomous vehicles.
    - Convolutional Neural Networks (CNNs) became the standard for image processing, while Recurrent Neural Networks (RNNs) improved sequential data analysis.
    - Companies like Google, Facebook, and Microsoft heavily invested in deep learning research.
7. **2020s: Continued Advancements**
    - Deep learning continues to evolve, with research focusing on more efficient training techniques, transfer learning, and reinforcement learning.
    - Generative Adversarial Networks (GANs) and Transformers, such as BERT and GPT models, gained prominence in natural language processing and generative tasks.

The history of ANNs is marked by periods of hype and disappointment, but recent decades have seen remarkable progress. Deep learning has become a fundamental component of artificial intelligence and has applications in a wide range of fields, from healthcare to finance to autonomous robotics. Researchers and engineers continue to explore new architectures and techniques to push the boundaries of what ANNs can achieve.

The overview of biological Neuron - systems and ANNs

The Overview of Biological Neuron System

The biological neuron system refers to the complex network of neurons (nerve cells) found in the human and animal nervous systems.

Humans have made several attempts to mimic the biological systems, and one of them is artificial neural networks inspired by the biological neural networks in living organisms. However, they are very much different in several ways. For example, the birds had inspired humans to create airplanes, and the four-legged animals inspired us to develop cars. The artificial counterparts are definitely more powerful and make our life better. The perceptrons, who are the predecessors of artificial neurons, were created to mimic certain parts of a biological neuron such as dendrite, axon, and cell body using mathematical models, electronics, and whatever limited information we have of biological neural networks.

Components and Working of Biological Neural Networks

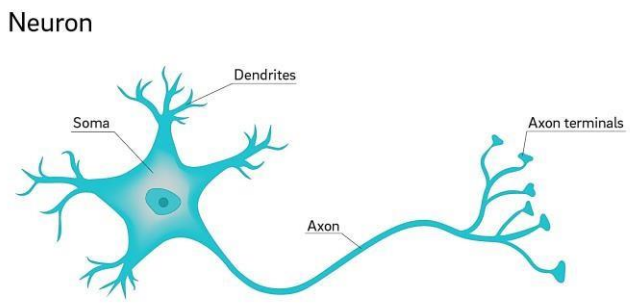

Figure 3 Parts of a biological neural network

In living organisms, the brain is the control unit of the neural network, and it has different subunits that take care of vision, senses, movement, and hearing. The brain is connected with a dense network of nerves to the rest of the body's sensors and actors. There are approximately 10^{11} neurons in the brain, and these are the building blocks of the complete central nervous system of the living body.

The neuron is the fundamental building block of neural networks. In the biological systems, a neuron is a cell just like any other cell of the body, which has a DNA code and is generated in the same way as the other cells. Though it might have different DNA, the function is similar in all the organisms. A neuron comprises three major parts: the cell body (also called Soma), the dendrites, and the axon. The dendrites are like fibers branched in different directions and are connected to many cells in that cluster.

Dendrites receive the signals from surrounding neurons, and the axon transmits the signal to the other neurons. At the ending terminal of the axon, the contact with the dendrite is made through a synapse. Axon is a long fiber that transports the output signal as electric impulses along its length. Each neuron has one axon. Axons pass impulses from one neuron to another like a domino effect.

Why Understand Biological Neural Networks?

For creating mathematical models for artificial neural networks, theoretical analysis of biological

neural networks is essential as they have a very close relationship. And this understanding of the brain's neural networks has opened horizons for the development of artificial neural network systems and adaptive systems designed to learn and adapt to the situations and inputs.

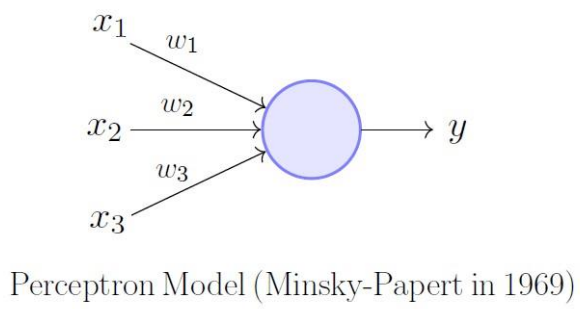

Figure 4 an artificial neuron

The Need of Biological Neural Network

Neural network, a network of simple processors (neurons) is found everywhere in the organism: in human brain, in every animal brain and in our heart, pancreas or lungs. It's a very efficient mechanism, whose functioning principle is based on the learning process that makes these systems very adaptive. The study of biological neural networks is important to understand and simulate the functioning of our own brain, the best known and most complex biological neural network in the world. This can lead to the development of bio-inspired artificial neural networks.

The Characteristics of a Biological Neural Network

A biological neural network is a network of neurons that are connected together by axons and dendrites. The connections between neurons are made by synapses. The axons transport chemicals that cause neurotransmitters to be released onto dendrites, where the neurotransmitters are then able to excite or inhibit an adjacent neuron. The neural network is able to learn and remember information, allowing it to solve problems or make decisions.

The overview of Artificial Neural Networks (ANNs) system

Introduction to Artificial Neural Networks

Artificial Neural Networks are the most popular machine learning algorithms today. The invention of these Neural Networks took place in the 1970s but they have achieved huge popularity due to the recent increase in computation power because of which they are now virtually everywhere. In every application that you use, Neural Networks power the intelligent interface that keeps you engaged.

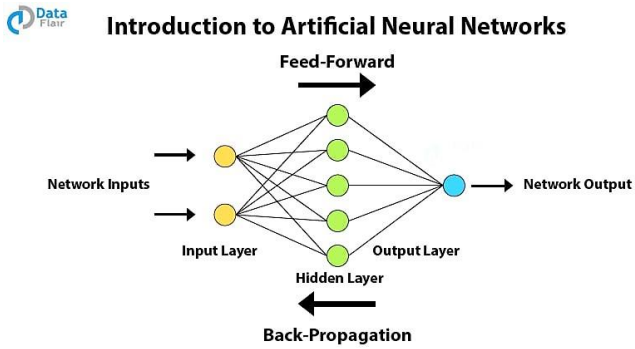

Figure 5 *ANN Diagram*

What Artificial Neural Network is:

Artificial Neural Networks are a special type of machine learning algorithms that are modelled after the human brain. That is, just like how the neurons in our nervous system are able to learn from the past data, similarly, the ANN is able to

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learn from the data and provide responses in the form of predictions or classifications.

ANNs are nonlinear statistical models which display a complex relationship between the inputs and outputs to discover a new pattern. A variety of tasks such as image recognition, speech recognition, machine translation as well as medical diagnosis makes use of these artificial neural networks.

An important advantage of ANN is the fact that it learns from the example data sets. Most commonly usage of ANN is that of a random function approximation. With these types of tools, one can have a cost-effective method of arriving at the solutions that define the distribution. ANN is also capable of taking sample data rather than the entire dataset to provide the output result. With ANNs, one can enhance existing data analysis techniques owing to their advanced predictive capabilities.

Types of Artificial Neural Networks

There are two important types of Artificial Neural Networks –

- FeedForward Neural Network
- FeedBack Neural Network

FeedForward Artificial Neural Networks

In the feedforward ANNs, the flow of information takes place only in one direction. That is, the flow of information is from the input layer to the hidden layer and finally to the output. There are no feedback loops present in this neural network. These type of neural networks are mostly used in **supervised learning** for instances such as classification, image recognition etc. We use them in cases where the data is not sequential in nature.

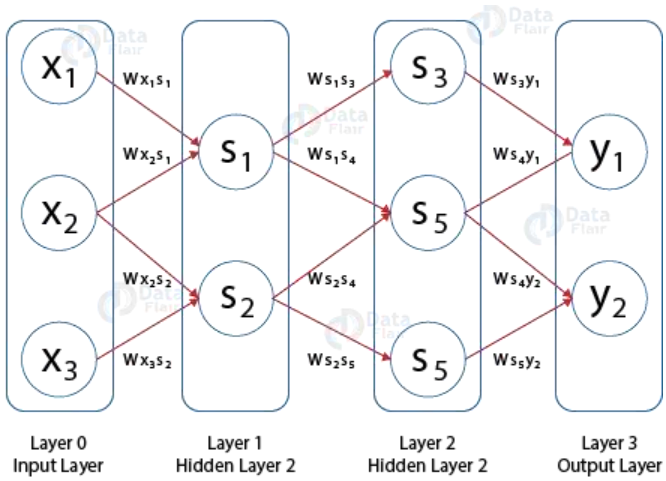

Figure 6 flow of information in FeedForward Neural

Network

Feedback Artificial Neural Networks

In the feedback ANNs, the feedback loops are a part of it. Such type of neural networks are mainly for memory retention such as in the case of recurrent neural networks. These types of networks are most suited for areas where the data is sequential or time-dependent.

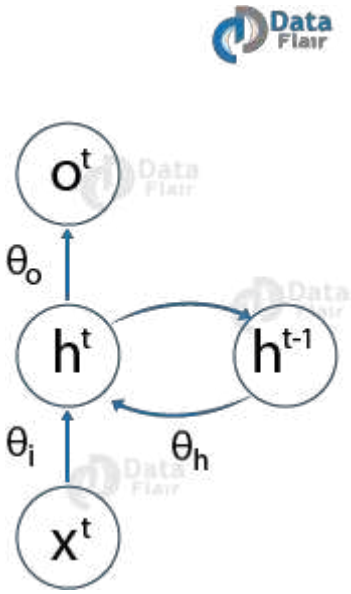

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Figure 7 Feedback in Artificial Neural Networks

Artificial Neural Networks Architecture

Introduction to the Architecture of ANN

Artificial neural network (ANN) architecture. ANNs consist of artificial neurons. Each artificial neuron has a processing node ('body') represented by circles in the figure as well as connections from ('dendrites') and connections to ('axons') other neurons which are represented as arrows in the figure. In a commonly used ANN architecture, the multilayer perceptron, the neurons are arranged in layers. An ordered set (a vector) of predictor variables is presented to the input layer. Each neuron of the input layer distributes its value to all of the neurons in the middle layer. Along each connection between input and middle neurons there is a connection weight so that the middle neuron receives the product of the value from the input neuron and the connection weight. Each neuron in the middle layer takes the sum of its weighted inputs and then applies a nonlinear (usually logistic) function to the sum. The result of the function then becomes the output from that particular middle neuron. Each middle neuron is connected to the output neuron. Along each connection between a middle neuron and the output neuron there is a connection weight. In the final step, the output neuron takes the weighted sum of its inputs and applies the non-linear function to the weighted sum.

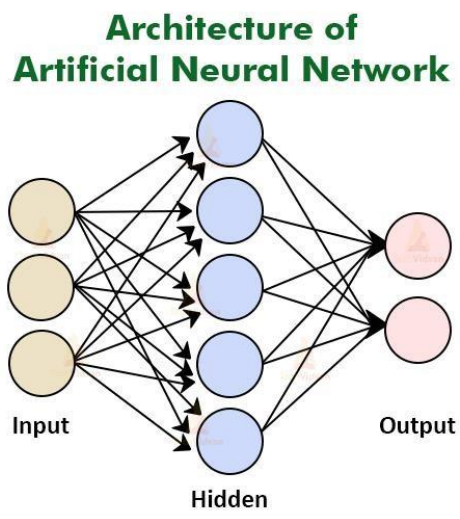

Figure 8 Architecture of ANN

Working of ANN Architecture

Firstly, the information is feed into the input layer. Which then transfers it to the hidden layers, and interconnection between these two layers assign weights to each input randomly at the initial point. Then bias is add to each input neuron and after this, the weight sum which is a combination of weights and bias is pass through the activation function. Activation Function has the responsibility of which node to fire for feature extraction and finally output is calculate. Therefore this whole process is known as Forward Propagation. After getting the output model to compare it with the original output and the error is known and finally, weights are updates in backward propagation to reduce the error and this process continues for a certain number of epochs (iteration). Finally, model weights get updates and prediction is done.

Input Layers

The *input layer* is the first layer of an ANN that receives the input information in the form of

various texts, numbers, audio files, image pixels, etc.

Hidden Layers

In the middle of the ANN model are the *hidden layers*. There can be a single hidden layer, as in the case of a perceptron or multiple hidden layers. These hidden layers perform various types of mathematical computation on the input data and recognize the patterns that are part of.

Output Layer

In the *output layer*, we obtain the result that we obtain through rigorous computations performed by the middle layer.

In a neural network, there are multiple parameters and hyperparameters that affect the performance of the model. The output of ANNs is mostly dependent on these parameters. Some of these parameters are weights, biases, learning rate, batch size etc. Each node in the ANN has some weight.

Each node in the network has some weights assigned to it. A transfer function is used for calculating the weighted sum of the inputs and the bias.

Back Propagation in Artificial Neural Networks

In order to train a neural network, we provide it with examples of input-output mappings. Finally, when the neural network completes the training, we test the neural network where we do not provide it with these mappings. The neural network predicts the output and we evaluate how correct the output is using the various error functions. Finally, based on the result, the model adjusts the weights of the neural networks to optimize the network following gradient descent through the chain rule.

Conclusion

The history of Artificial Neural Networks (ANNs) has witnessed significant evolution, stemming from inspiration drawn from biological neural networks. Despite its early inception in the 1940s and 1950s, ANNs faced challenges and periods of stagnation. However, with the advent of increased computing power and the introduction of backpropagation in the 1980s, ANNs experienced a resurgence, becoming pivotal in machine learning and artificial intelligence. Comparing ANNs to biological neurons, while they share fundamental concepts, ANNs are simplified models optimized for information processing. ANNs' architecture, characterized by layered structures with interconnected neurons and weighted connections, empowers them to excel in pattern recognition and complex data processing tasks, contributing significantly to various real-world applications. Understanding this architecture is vital for effective design and training of neural network models.

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